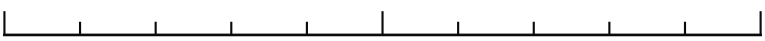
# Lidarbot print sheets

Print every page at 100% ("actual size"; never "fit to page"), on matte paper or matte card.
Glossy paper reflects the overhead lights and the camera loses the marker.
Check the 100 mm bar at the bottom of each page with a ruler before using anything.
Keep the white border around each marker (cut on the dashed line, not at the black edge).
Mount markers flat: on card or foam board, no bubbles or curl.

| Page | What | Size | Use |
|---|---|---|---|
| 2 | Robot marker ID 0 (+ spare) | 60 mm | robot top plate, arrow to the front |
| 3-4 | Floor markers IDs 1-4 | 80 mm | tape to the floor at measured spots |
| 5 | Wall target, IDs 10-13 | 180 mm board | wall ahead of the tape start |
| 6 | Servo bench markers IDs 20-23 | 35 mm | on a disc or arm on each servo horn |
| 7 | Checkerboard 10 x 7 | 25 mm squares | camera intrinsics |

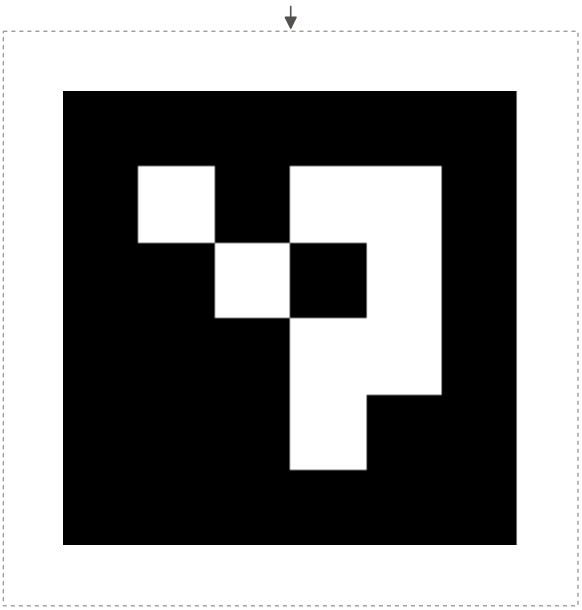
After printing, measure and write down the real black-square size of each marker type.
If it differs from the nominal size by more than 0.5 mm, put the measured value in the config:
lab/lab.json (robot_marker.size_m, floor_marker_size_m) or tools/wall_target.py (MARKER_MM).



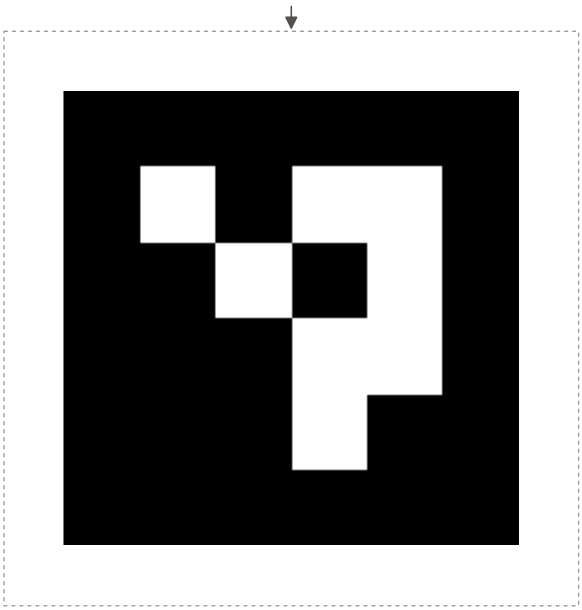
must measure 100 mm. If not, reprint at 100% / actual size

# Robot marker (overhead tracker)

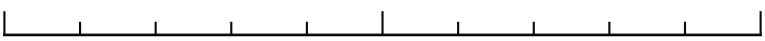
Tape one to the robot's top plate, flat, with the arrow pointing toward the robot's front.
Its height above the floor goes in lab.json (robot_marker.height_m). The second copy is a spare.



ID 0 - 60 mm (robot)



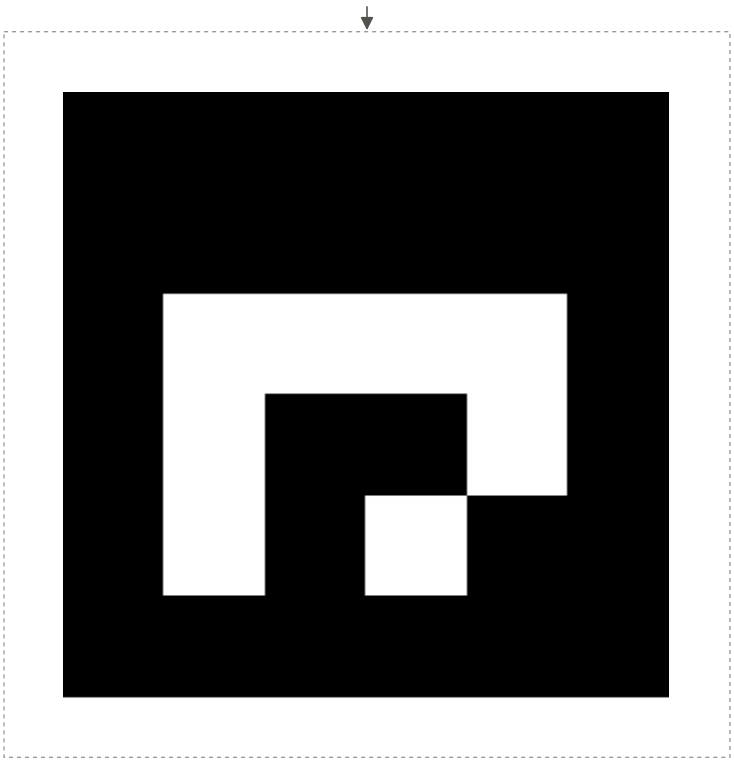
ID 0 - 60 mm (spare)



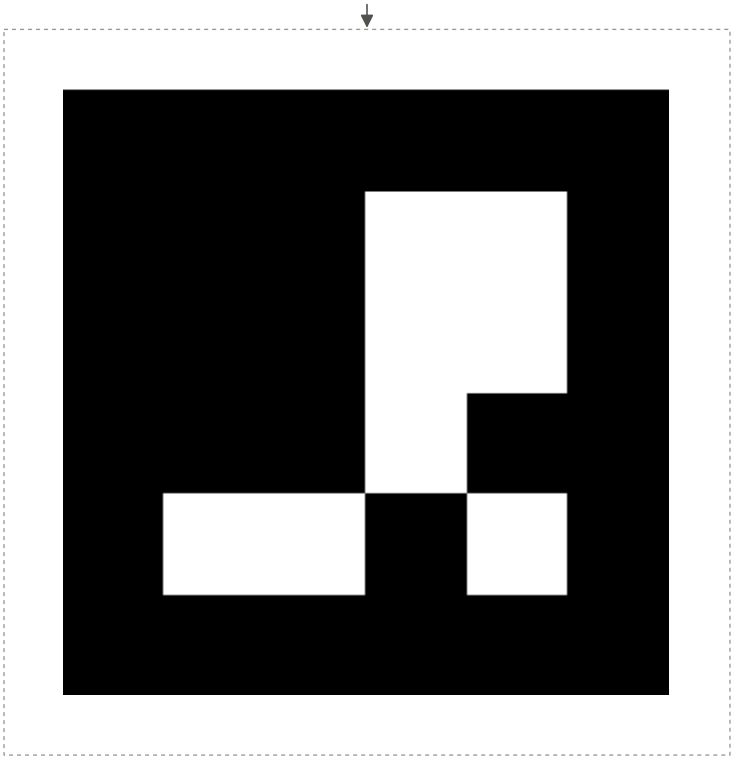
must measure 100 mm. If not, reprint at 100% / actual size

# Floor markers 1 and 2 (overhead tracker)

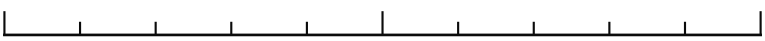
Tape flat to the floor at tape-measured positions around the test area (lab.json floor_markers).
Measure each marker's center from the tape start. Arrow points along the tape (+x).



ID 1 - 80 mm



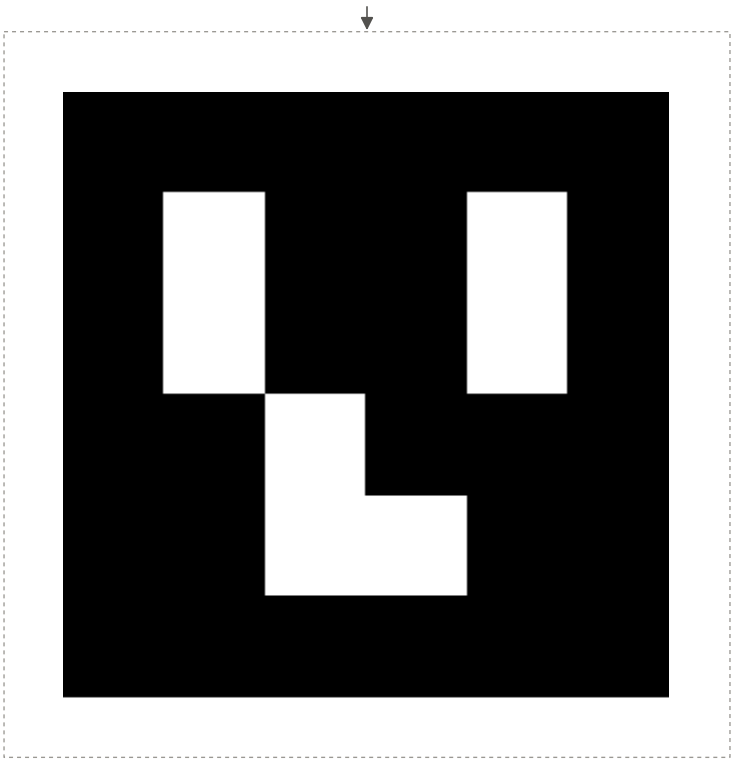
ID 2 - 80 mm



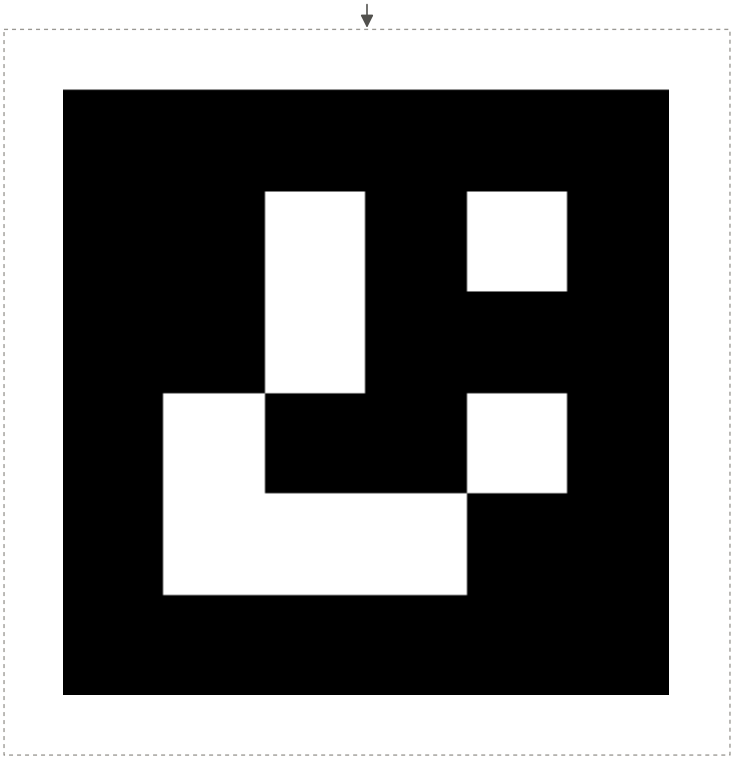
must measure 100 mm. If not, reprint at 100% / actual size

# Floor markers 3 and 4 (overhead tracker)

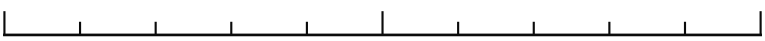
Tape flat to the floor at tape-measured positions around the test area (lab.json floor_markers).
Measure each marker's center from the tape start. Arrow points along the tape (+x).



ID 3 - 80 mm



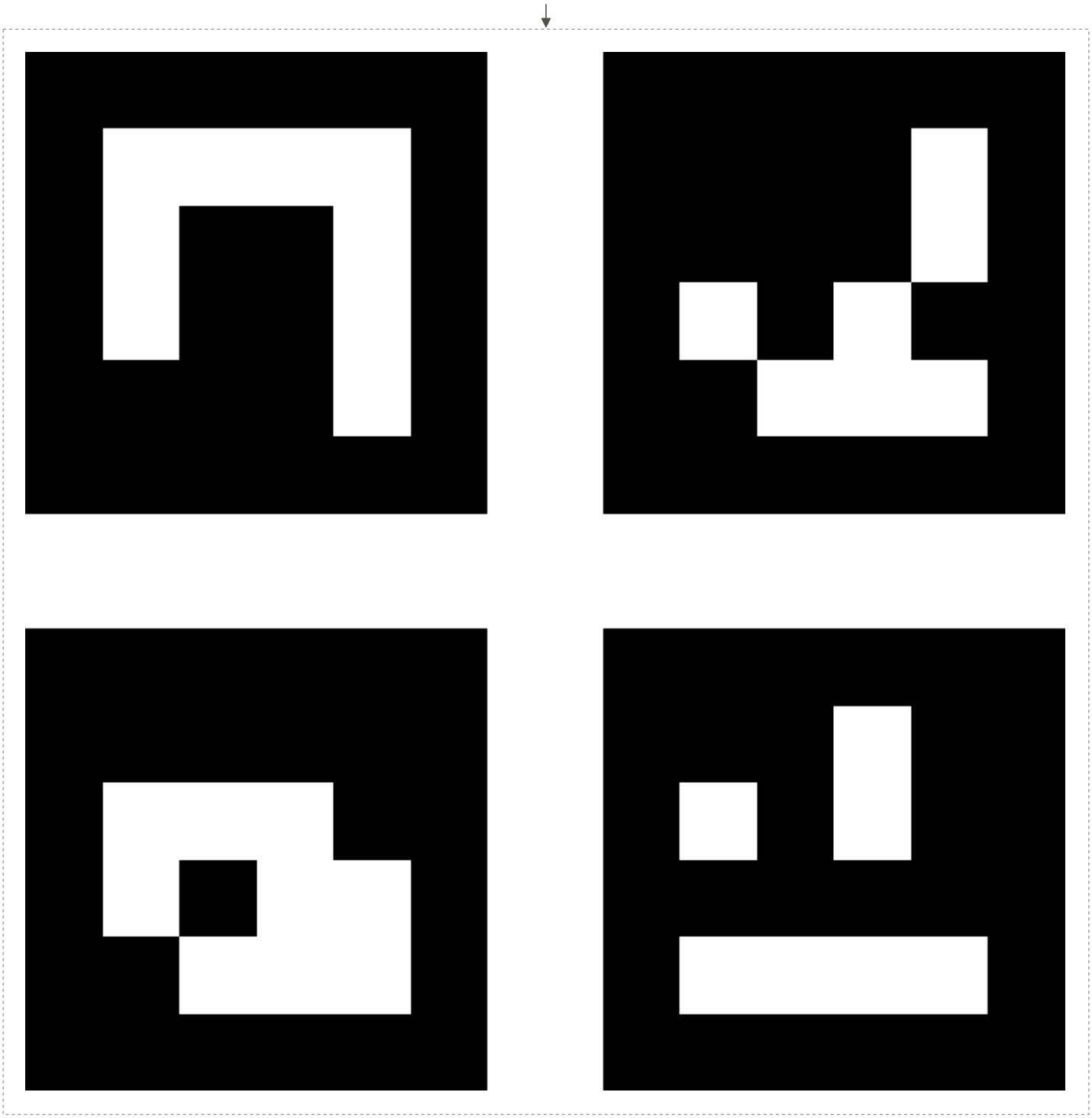
ID 4 - 80 mm



must measure 100 mm. If not, reprint at 100% / actual size

# Wall target (camera drift check)

Mount flat on the wall ahead of the tape start, centered at the robot camera's height, arrow up.
Four 80 mm markers with 20 mm gaps; tools/wall_target.py expects exactly this layout.

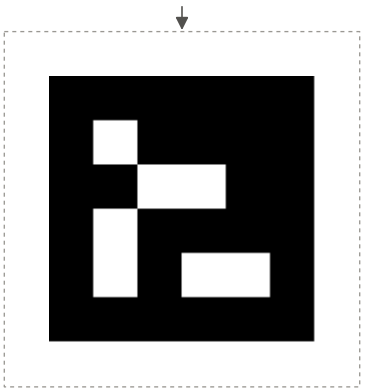


IDs 10 (top left), 11 (top right), 12 (bottom left), 13 (bottom right)

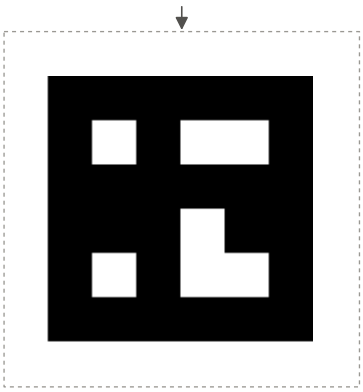
must measure 100 mm. If not, reprint at 100% / actual size

# Servo test bench markers

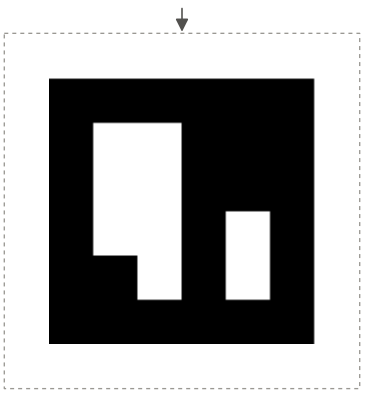
Glue one to a light disc or arm on each servo horn, centered on the output shaft, flat.
One marker per servo under test; the overhead camera reads its angle.



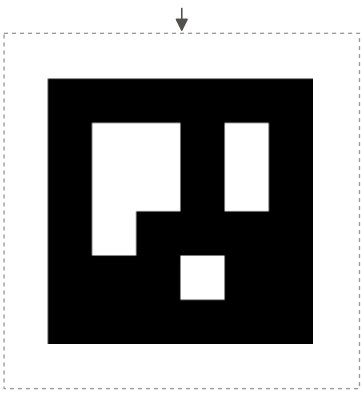
ID 20 - 35 mm



ID 21 - 35 mm



ID 22 - 35 mm

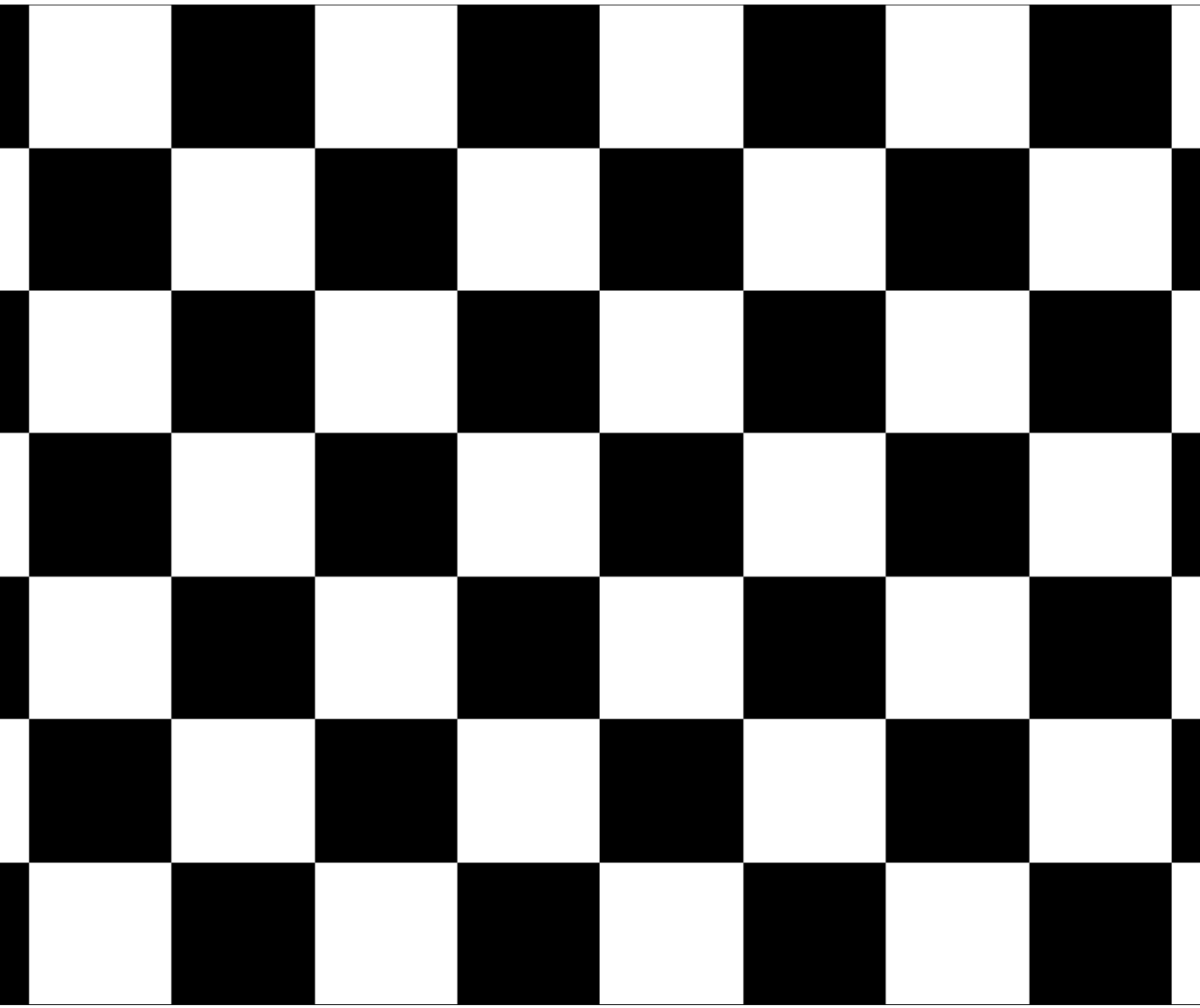


ID 23 - 35 mm

must measure 100 mm. If not, reprint at 100% / actual size

# Camera calibration checkerboard

10 x 7 squares of 25 mm (9 x 6 inner corners). Mount on stiff board,
perfectly flat. Show it to the camera at many angles and distances (OpenCV calibrateCamera).



must measure 100 mm. If not, reprint at 100% / actual size